

Quarter-Root-Only Ablation of the Two-Regime Penetration Model

Why dropping $k_{\sqrt{t}}$ improves the fit

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- **Background:** the original two-regime model and what it stores.
- **Motivation:** the $\sqrt{t}$ branch contributes essentially nothing in successful fits.
- **Ablation:** replace by a single quarter-root branch with sigmoid onset.
- **Results:** success rate +28.6% absolute; flagged-set median RMSE drops from 8.12 mm to 1.08 mm.

Original Two-Regime Model

Each plume penetration $S(t)$ is fitted by

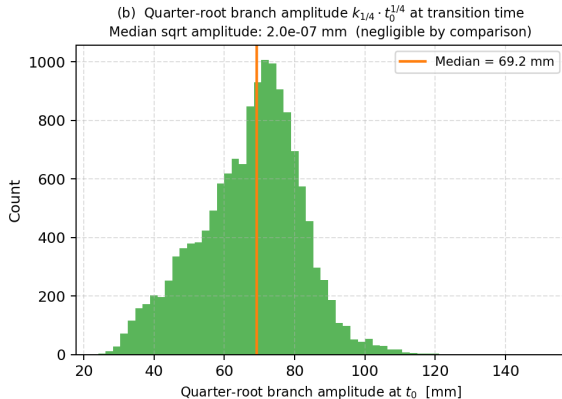
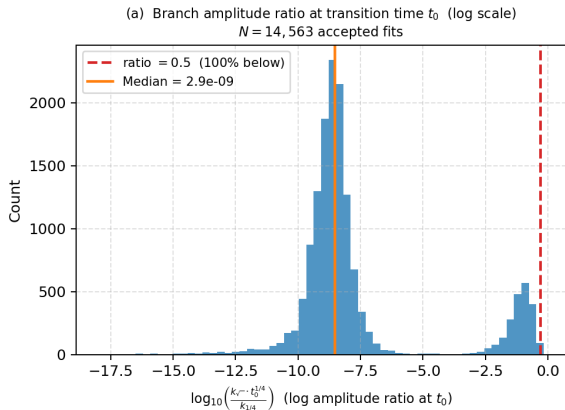
$$S(t) = (1 - w(t)) k_{\sqrt{}} \sqrt{t} + w(t) k_{1/4} t^{1/4}, \quad w(t) = \sigma\left(\frac{t - t_0}{s}\right)$$

Physical interpretation:

- **Early phase** ($t < t_0$): momentum-driven free jet $\Rightarrow \sqrt{t}$ (Naber–Siebers).
- **Late phase** ($t > t_0$): impingement / deceleration $\Rightarrow t^{1/4}$.
- Sigmoid $w(t)$ blends the two regimes around transition time t_0 .

Stored constants per curve (log-space): $\log k_{\sqrt{}}$, $\log k_{1/4}$, $\log t_0$, $\log s$ (4 parameters).

Empirical Observation: $\sqrt{t}$ branch is negligible



Reading the Distributions

Left panel: amplitude ratio at the transition time t_0

$$r = \frac{k_{\sqrt{}} \sqrt{t_0}}{k_{1/4} t_0^{1/4}}$$

- Median $r \approx 2.9 \times 10^{-9}$.
- **100% of accepted fits** have $r < 0.5$.
- Bulk of mass sits 8–10 decades *below* the quarter branch.

Right panel: actual amplitudes at t_0

- Quarter branch median: $k_{1/4} t_0^{1/4} \approx 69$ mm.
- Sqrt branch median: $\approx 2 \times 10^{-7}$ mm (numerically zero).

Implication. The solver consistently drives $k_{\sqrt{}}$ to a degenerate small value: the early $\sqrt{t}$ regime is not actually present in the truncated, impingement-dominated penetration data we collect.

Why this Matters for Failed Fits

The 4-parameter model has too many degrees of freedom relative to what the data constrains:

- On clean (well-behaved) traces, the solver still converges, but with $k_{\sqrt{}}$ effectively dead.
- On flagged / hard traces, $k_{\sqrt{}}$ becomes a destructive degree of freedom: it inflates during fitting and **dominates the extrapolation** to 5 ms used by the masking pipeline.
- Result: many physically reasonable curves are killed by the `penetration_far` check, even when the observed window is well-fitted.

Hypothesis: a 3-parameter quarter-only model with sigmoid onset will match the same window quality on clean data, while extrapolating safely on flagged data.

Quarter-Only Ablation Model (q1)

Replace the blend by a single regime gated by a sigmoid onset:

$$S_{q1}(t) = \sigma\left(\frac{t - t_0}{s}\right) k_{1/4} t^{1/4}$$

Stored constants: $\log k_{1/4}$, $\log t_0$, $\log s$ (3 parameters).

Implementation. Run alongside the original model in `fit_raw_data.py`:

- `ABLATION_QUARTER_ONLY = True` fits both models per series.
- Same Huber-loss `least_squares` solver, fresh initial guess.
- Equivalent masking pipeline applied to q1 (`flag_bad_fit_q1`): finite-param checks, t_0 , s bounds, far-time penetration in $[18, 300]$ mm, and grouped robust-z outlier detection.
- Success criterion: $\neg \text{flag_bad_fit} \wedge \text{RMSE} < 3 \text{ mm}$ (symmetric across both models).

Aggregate Success Rate (266 folders, 23,495 series)

	Main (4-param)	q1 (3-param)
Successful series	14,406	21,142
Success rate	61.3%	90.0%
Absolute gain	+28.6 percentage points	

Per-nozzle gains (q1 minus main, percentage points):

- DS300 (highly impinged): +37.6
- Nozzle 7: +28.7
- Nozzle 3 (after MIN_SERIES_POINTS filter): +22.4
- Nozzle 5: +19.6
- Nozzles 1, 2, 4, 6, 8: +10 to +14

No nozzle regressed. The improvement is largest where impingement dominates — consistent with the physical motivation.

RMSE: Where the Real Win Lies

Median RMSE	Main	q1
Clean subset	1.23 mm	1.27 mm
Flagged subset	8.12 mm	1.08 mm

Two readings:

- **On clean traces** the two models are statistically indistinguishable — removing $k_{\sqrt{}}$ costs nothing.
- **On flagged traces** q1 reduces median residual by $\approx 7.5\times$. The flagged set becomes *usefully fitted* rather than degenerately extrapolated.

Visual Diagnostic (typical flagged folder)

On flagged plots the two models are visually distinguishable:

- **Main (dashed)** curves diverge to 100–130 mm by 2.5 ms, driven by an inflated $\sqrt{t}$ tail outside the observation window.
- **q1 (dotted)** curves track the data within the window and plateau under 100 mm at long times — the bounded $t^{1/4}$ behavior the masking pipeline expects.

This explains the asymmetry quantified above: same data, better extrapolation behavior, more series surviving the physical-plausibility gate.

Conclusion and Next Steps

Take-aways

- The $k_{\sqrt{t}}$ branch is empirically dead in our data (median ratio $\sim 10^{-9}$, 100% below 0.5).
- Removing it removes a destructive degree of freedom on hard traces.
- Net effect: +28.6 pp success rate, $\approx 7.5\times$ smaller median residual on the flagged subset, and one fewer parameter to store and to learn downstream.

Open questions

- Should q1 become the default model for the MLP target, with the 4-param fit retained only as diagnostic baseline?
- Re-derive Stage-1 / Stage-2 features from $(\log k_{1/4}, \log t_0, \log s)$ and check whether feature instability also drops.